In Silico Closed-Loop Metabolic Gating of Glycolytic-to-Oxidative Transition in *Ex Vivo* CAR-T Perfusion Bioreactors Guided by Structural Bioinformatics and Optical Redox Observability

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November 2026 (MitoRestorer Series, Paper II; Computational Study — MIASE Seed: 20261111)

Abstract

Durable clinical responses in adoptive chimeric antigen receptor (CAR) T-cell therapy correlate with infusing stem-like memory T cells (T_{SCM}, $CD45RA^+CD62L^+CCR7^+TCF7^+$) possessing high mitochondrial spare respiratory capacity (SRC), yet 168-hour *ex vivo* perfusion expansion under continuous high glucose and interleukin-2 (IL-2) enforces a Warburg glycolytic program that accumulates depolarized mitochondria and TOX^+ epigenetic exhaustion. Here we present a strictly *in silico* systems-and-control investigation—combining structural bioinformatics, eight-state nonlinear ordinary differential equation (ODE) modeling, discrete-adjoint optimal control, physical inline sensor observability analysis, and freestanding fixed-point firmware verification—to design falsifiable closed-loop metabolic gating protocols prior to wet-lab execution. Heavy-atom analysis of experimental crystal structures confirms druggable allosteric pockets on pyruvate dehydrogenase kinase 1 (PDK1, PDB: 2Q8G, 1.90 Å; lipoyl-pocket Ser75 hydrogen bond at 2.50 Å to AZD7545) and sirtuin 1 (SIRT1, PDB: 4ZZJ, 2.74 Å; Glu230 allosteric hinge at 3.47 Å), whereas full-length AlphaFold v6 models of downstream effectors FOXO1 (655 AA, mean pLDDT = 50.96) and PGC-1α (798 AA, mean pLDDT = 52.74) exhibit single structured nucleic-acid-binding modules flanked by intrinsically disordered post-translational modification regions (9/9 Akt/SIRT1 sites on FOXO1 and 39/47 lysines on PGC-1α have pLDDT < 70). Building on Paper I, our simulated two-stage **Bioreactor-BCR** controller separates Stage-1 mitophagy-assisted glycolytic blastogenesis (u_1) from Stage-2 pyruvate dehydrogenase reactivation and NAD^+/SIRT1-coupled oxidative consolidation (u_2), yielding 62.80× fold expansion, 73.20% T_{SCM}, SRC = 4.87×, and $h(168\text{ h}) = 6.73\%$ on the nominal model, closely tracking the 200-variable discrete-adjoint Pontryagin Pareto frontier (all-200-variable gradient error 2.04×10^{-10}, 5-start cost spread 1.12%). Across $N = 1{,}000$ simulated heterogeneous oncology apheresis donors with unmodeled kinetic dispersion and ±6% optical-density (OD) lymphoblast-size calibration bias, physical 4-channel sensing (Lac:Pyr, volumetric OUR_{vol}, OD, and pH) fails (40.5% 5-criteria pass rate) because cell-density bias and Warburg-rate heterogeneity confound specific respiration. Augmenting the observer with a ratiometric autofluorescence optical redox ratio (ORR)—in which cell concentration cancels analytically—restores state observability ($\hat{h}$ RMSE 2.08%), lifting the 168-hour pass rate from 54.1% (best fixed 68 h timer) to 58.7% (paired McNemar $p = 8.04 \times 10^{-4}$; vs. 4-channel $p = 1.19 \times 10^{-41}$), retaining 59.4% under 100% enzymatic Lac:Pyr biofouling via NIS > 13.5 optical handover ($p = 5.22 \times 10^{-129}$ vs. 12.3% without ORR), and reaching 74.0% ($p = 2.08 \times 10^{-33}$) when paired with OD-triggered adaptive harvest (168–240 h) for hypoproliferative donors. Finally, freestanding FPU-free C99 firmware implementing

both the Q16.16/Q8.24 ΣΔ RK4 plant (0.0078%–0.0114% error) and the 5-channel sequential Huber-EKF (0.0216% parity vs. FP64) executes within a zero-heap 2,704 B (33.01%) static SRAM workspace and rescues simulated perfusion-line acidosis (min pH 6.14 → 6.91).

1 Introduction

1.1 Bioenergetic Determinants of CAR-T Persistence

In autologous CD19- and BCMA-directed chimeric antigen receptor (CAR) T-cell manufacturing, clinical durability in chronic lymphocytic leukemia, B-cell acute lymphoblastic leukemia, and non-Hodgkin lymphoma is governed less by total infused cell count than by the proportion of early stem-cell memory T lymphocytes (T_{SCM}, $CD8^+CD45RA^+CD62L^+CCR7^+TCF7^+$) and their mitochondrial spare respiratory capacity (SRC) [1, 2]. Transcriptomic and epigenetic studies have established two converging mechanistic axes underlying this phenotype: first, nuclear retention of Forkhead box protein O1 (FOXO1) acts as an essential master regulator of CAR-T memory programming and metabolic fitness while opposing TOX^+ terminal exhaustion [3, 4, 5]; second, selective clearance of depolarized, fragmented mitochondria via PINK1/Parkin-dependent mitophagy (stimulated *ex vivo* or *in vivo* by urolithin A) preserves fused cristae networks and expands T_{SCM} subsets [6, 7, 8].

1.2 The Ex Vivo Proliferation–Exhaustion Conflict

Automated closed-cassette perfusion bioreactors (such as hollow-fiber or gas-permeable membrane cassettes) activate patient apheresis T cells via anti-CD3/CD28 stimulation over a 7-day (168 h) expansion cycle. This manufacturing window imposes a well-known dynamical trade-off between clonal blastogenesis and bioenergetic stemness [9, 2]:

1. **Unphased Aerobic Glycolysis (Warburg Lock):** Continuous exposure to high glucose (25 mM) and high-dose IL-2 (300 IU/mL) drives mTORC1, Akt, and HIF-1α signaling. HIF-1α upregulates pyruvate dehydrogenase kinase 1 (PDK1) [10], which phosphorylates the pyruvate dehydrogenase (PDH) E1α subunit, shunting pyruvate to lactate (Lac : Pyr $>$ 25) and acidifying the perfusate. Concurrently, Akt phosphorylates FOXO1 at Thr24, Ser256, and Ser319, triggering 14-3-3 chaperone binding and nuclear export [3, 4], while mTORC1 suppresses ULK1-initiated mitophagy, allowing depolarized organelles (D) and TOX^+ chromatin remodeling to accumulate [9, 5].
2. **Premature Oxidative Forced Switching (Dose Failure):** Conversely, forcing oxidative phosphorylation or restricting glycolytic flux immediately upon CD3/CD28 activation ($t = 0$ h) deprives dividing lymphoblasts of glycolytic ribonucleotide, serine, and lipid precursors required during early log-phase expansion, risking failure to reach minimum clinical release doses ($\geq 50\times$ viable fold expansion).
3. **Oncology Apheresis Heterogeneity and Sensor Drift:** Unlike healthy-donor research runs, heavily pretreated oncology patients enter apheresis with wide baseline dispersion in T-cell mitochondrial depolarization ($h_0 \in [0.20, 0.62]$) and division kinetics [1]. Consequently, a fixed open-loop media-switch timer tuned to a nominal donor risks switching slow-clearing donors prematurely or holding rapid-proliferating donors in prolonged glycolysis. Yet closed-loop gating using inline enzymatic supernatant sensors faces its own bioprocess hazard: amperometric lactate/pyruvate electrodes undergo progressive protein biofouling over multi-day perfusion.

1.3 Epistemic Scope and Computational Contributions

Disclosure of Computational Scope: This study is strictly a theoretical, structural-bioinformatic, and *in silico* control-engineering investigation. *No wet-lab, in vitro, ex vivo, or in vivo biological experiments were performed.* All cell-expansion trajectories, heterogeneous patient-cohort statistics, biofouling fault responses, and embedded processor benchmarks are generated by deterministic mathematical models and host-emulated firmware designed to formulate quantitative, falsifiable predictions for subsequent experimental testing (§7).

Extending the nondimensional staged heteroplasmy control theory of Paper I [11] to proliferating cell populations coupled to extracellular bioreactor mass transfer, this paper makes four specific contributions:

1. **Unified 8-State Perfusion Cassette ODE with 1-to-1 Code Parity (§2 and Appendix A.1):** We formulate an eight-state bioreactor model coupling intracellular mitochondrial subpopulations (W, D), clonal cell fold expansion (N_{cells}), supernatant lactate-to-pyruvate ratio (R_{LP}), nuclear FOXO1 fraction (F_{nuc}), cristae-linked SRC (S_{SRC}), epigenetic exhaustion (E_{TOX}), and perfusate pH, with every closed-form auxiliary equation verified identically across FP64 Python, smooth discrete-adjoint optimization, and bare-metal C99 fixed-point code.
2. **Atomic Contact & AlphaFold v6 Disorder Audit Guiding Actuator Selection (§3):** By auditing heavy-atom ligand contacts in PDB: 2Q8G (PDK1–AZD7545) and PDB: 4ZZJ (SIRT1–STAC–carba-NAD$^+$) alongside AlphaFold Database v6 pLDDT and Predicted Aligned Error (PAE) matrices for full-length FOXO1 and PGC-1α, we quantify why disordered regulatory PTM tails lack preformed small-molecule pockets and why upstream enzymatic actuation—paired with explicit controls for known STAC chemical-probe artifacts [12, 13]—provides a structurally grounded intervention strategy.
3. **Discrete-Adjoint Pareto Frontier, Ratiometric ORR Observability & Extended-Horizon Rescue (§4–§5):** Solving a 200-variable discrete-adjoint optimal control problem (L-BFGS-B, all-200-variable gradient error 2.04×10^{-10}, 5-start spread 1.12%) maps the proliferation–stemness Pareto frontier. Across $N = 1{,}000$ heterogeneous oncology donors under physical volumetric OUR_{vol} and ±6% OD scattering bias, we prove why cell concentration cancels out identically in ratiometric autofluorescence ORR [14], lifting the 168-hour pass rate from 40.5% (4-channel No-ORR) and 54.1% (best fixed 68 h timer) to 58.7% (exact McNemar $p = 8.04 \times 10^{-4}$), retaining 59.4% under 100% Lac:Pyr biofouling via NIS > 13.5 optical handover ($p = 5.22 \times 10^{-129}$ vs. 12.3% without ORR), and reaching 74.0% ($p = 2.08 \times 10^{-33}$) under OD-guided adaptive harvest (168–240 h).
4. **Plant-Preserving Acidosis Flush & Deterministic FPU-Free C99 Safety Firmware (§6):** Eliminating unphysical state-reset assumptions, we verify a continuous-plant neutral-buffer bypass flush (u_{fl}) during simulated perfusion-line occlusion and demonstrate freestanding FPU-free C99 execution of both the Q16.16/Q8.24 ΣΔ RK4 integrator and the 5-channel sequential Huber-EKF (0.0216% parity vs. FP64) within a zero-heap 2,704 B static SRAM workspace.

2 Mathematical Model of the 168-Hour Perfusion Cassette

2.1 State Variables, Bounded Cassette Actuators, and Core Bioenergetic Balance

We model an automated closed-cassette CAR-T perfusion culture over $t \in [0, T_{\max}]$ ($T_{\max} = 168.0$ h) via the eight-dimensional state vector $\mathbf{x}(t) = [W, D, N_{\text{cells}}, R_{\text{LP}}, F_{\text{nuc}}, S_{\text{SRC}}, E_{\text{TOX}}, \text{pH}]^\top$. Here $W(t) \geq 0$ denotes normalized intact, fused-cristae wild-type mitochondrial mass per cell; $D(t) \geq 0$ is normalized depolarized/fragmented mitochondrial mass per cell; $N_{\text{cells}}(t) \geq 1$ is viable clonal fold expansion ($N_{\text{cells}}(0) = 1.0$);

$R_{\mathrm{LP}}(t) > 0$ is the extracellular supernatant molar [Lactate]/[Pyruvate] ratio; $F_{\mathrm{nuc}}(t) \in [0,1]$ is the nuclear-retained, transcriptionally active fraction of `FOXO1`; $S_{\mathrm{SRC}}(t) \geq 0$ is mitochondrial spare respiratory capacity (dimensionless multiple of basal oxygen consumption); $E_{\mathrm{TOX}}(t) \in [0,1]$ is the fraction of cells committed to TOX$^+$PD-1$^+$TIM-3$^+$ epigenetic exhaustion; and $\mathrm{pH}(t)$ is the cassette perfusate pH. Total normalized organelle mass is $M(t) = W(t) + D(t)$, with intracellular depolarized fraction (functional heteroplasmy) $h(t) = D(t)/M(t)$ and intact fraction $w_{\mathrm{frac}}(t) = 1 - h(t) = W(t)/M(t)$.

The cassette is driven by three bounded perfusate controls $\mathbf{u}(t) = [u_1(t), u_2(t), u_{\mathrm{fl}}(t)]^\top \in [0,1]^3$:

- $u_1(t) \in [0,1]$: **Stage-1 Mitophagy & Mild Glycolytic Restraint Perfusate (Media-A)**, modeling urolithin A (25 μM) mitophagy induction paired with moderate `PI3K`δ`/Akt` restraint during high-glucose blastogenesis [7, 4];
- $u_2(t) \in [0,1]$: **Stage-2 Oxidative Stemness Perfusate (Media-B)**, modeling iso-osmotic partial galactose substitution ($\alpha_{\mathrm{gal}} = 0.74$), allosteric `PDK1` lipoyl-pocket inhibition (`AZD7545`), NAD$^+$ repletion / `SIRT1` activation, and homeostatic IL-7/IL-15 cytokine support [10, 15];
- $u_{\mathrm{fl}}(t) \in [0,1]$: **Emergency Neutral-Buffer Bypass Flush Valve**, routing fresh $\mathrm{pH}_{\mathrm{media}} = 7.36$ buffer across the hollow-fiber cartridge if main perfusion-line occlusion triggers acute lactic acidosis.

Let $(z)_+ \equiv \max(0, z)$ denote the positive-part operator and $d_{\mathrm{ros}}(\mathbf{x}, \mathbf{u}) = k_{\mathrm{dmg}}(\mathrm{EP}(\mathbf{x}, \mathbf{u}) - \theta_{\mathrm{GSH}})_+$ denote threshold reactive-oxygen-species conversion of intact into depolarized organelles. The core intracellular organelle competition and clonal lymphoblast expansion dynamics take the form:

$$\dot{W} = \big(k_{\mathrm{bio},0} + (k_{\mathrm{bio,max}} - k_{\mathrm{bio},0})\, u_2 F_{\mathrm{nuc}}\big) W \Big(1 - \frac{M}{K}\Big)_+ - \big(k_{\mathrm{mito},W} + d_{\mathrm{ros}}(\mathbf{x}, \mathbf{u})\big) W, \tag{1a}$$

$$\dot{D} = k_{\mathrm{rep,adv}}\big(k_{\mathrm{bio},0} + \xi_D (k_{\mathrm{bio,max}} - k_{\mathrm{bio},0})\, u_2\big) D \Big(1 - \frac{M}{K}\Big)_+ - k_{\mathrm{mito},D}(u_1, R_{\mathrm{LP}})\, D + d_{\mathrm{ros}}(\mathbf{x}, \mathbf{u})\, W, \tag{1b}$$

$$\dot{N}_{\mathrm{cells}} = \Big[\mu_{\mathrm{glyc}} g_{\mathrm{glc}}(u_2)\big(1 - 0.08 u_1 (1 - u_2)\big) + \mu_{\mathrm{ox}} u_2 g_{\mathrm{pdh}}(u_2) \Phi_{\mathrm{ox}}(\mathbf{x})\Big] \phi_{\mathrm{pH}}(\mathrm{pH})\, \frac{\mathrm{ATP}_{\mathrm{tot}}}{\mathrm{ATP}_{\mathrm{tot}} + K_{\mathrm{ATP}}}\, N_{\mathrm{cells}} \Big(1 - \frac{N_{\mathrm{cells}}}{N_{\mathrm{cap}}}\Big)_+ . \tag{1c}$$

In (1), high extracellular lactate-to-pyruvate ratios R_{LP} partially mask `ULK1/PINK1` mitophagy through $k_{\mathrm{mito},D}(u_1, R_{\mathrm{LP}})$, extracellular acidosis below pH 7.02 suppresses division and nuclear import via $\phi_{\mathrm{pH}}(\mathrm{pH})$, and Stage-2 oxidative division $\mu_{\mathrm{ox}} u_2 g_{\mathrm{pdh}}(u_2) \Phi_{\mathrm{ox}}(\mathbf{x})$ requires both reactivated pyruvate dehydrogenase flux $g_{\mathrm{pdh}}(u_2)$ and intact cristae mass $w_{\mathrm{func}} = W(1 - h)$. The complete eight-state ODE system—including the coupled rate equations for $\dot{R}_{\mathrm{LP}}$, $\dot{F}_{\mathrm{nuc}}$, $\dot{S}_{\mathrm{SRC}}$, $\dot{E}_{\mathrm{TOX}}$, and $\dot{\mathrm{pH}}$ together with all ten closed-form auxiliary bioenergetic definitions ensuring 1-to-1 parity between `eval_cart8_rhs_vec` (Python) and `eval_cart8_rhs_q16_to_q24` (C99)—is given in Appendix A.1.

At harvest, the stem-like memory CAR-T fraction (%) is evaluated via the phenotypic readout

$$\mathrm{T}_{\mathrm{SCM}}(\mathbf{x}) = 100\, F_{\mathrm{nuc}}\, (1 - E_{\mathrm{TOX}})\, \frac{S_{\mathrm{SRC}}}{S_{\mathrm{SRC}} + K_{\mathrm{SRC,mem}}}, \tag{2}$$

capturing the joint requirement of nuclear `FOXO1` retention (F_{nuc}), absence of TOX$^+$ epigenetic scarring ($1 - E_{\mathrm{TOX}}$), and cristae bioenergetic reserve (S_{SRC}) [3, 4, 2].

2.2 Exact Stage-1 Quotient Selection and Co-Actuation Coupling

Proposition 1 (Stage-1 Quotient Selection)**.** *During Stage-1 mitophagy-primed expansion ($u_1 = 1, u_2 = 0$), whenever* $\mathrm{EP}(\mathbf{x}, \mathbf{u}) \leq \theta_{\mathrm{GSH}}$ *so* $d_{\mathrm{ros}} = 0$*, differentiating* $h = D/(W + D)$ *along* (1a)–(1b) *yields the exact Riccati*

Table 1: **Complete Parameterization and Provenance of the 8-State *In Silico* CAR-T Perfusion Bioreactor Model.** Every value matches `NOMINAL_PARAMS` in `simulator/generate_paper2_verified_benchmarks.py` and `build/coralnpu/cart_bioreactor_hil_testbench.c` 1-to-1. Tags: **[M]** primary literature scale; **[F]** calibrated phenomenological coupling; **[A]** operational bioreactor/cassette specification.

Subsystem	Parameter(s)	Nominal Value(s)	Units	Tag	Biophysical Role & Literature Basis
Mitochondrial (W, D)	$K,\ k_{\mathrm{mito},W},\ k_{\mathrm{bio},0}$	1.40, 0.008, 0.028	norm., h^{-1}, h^{-1}	[M/F]	Organelle cap & basal turnover [2, 11]
Mitophagy & Biogen.	$k_{\mathrm{mito},D,\max},\ k_{\mathrm{bio},\max},\ k_{\mathrm{rep,adv}},\ \xi_D$	0.068, 0.086, 1.22, 1.00	h^{-1}, h^{-1}, –, –	[F]	Urolithin A clearance & replicative bias [7, 16]
Warburg Masking	$R_{\mathrm{basal}},\ K_{\mathrm{LP}},\ \eta_D$	8.0, 22.0, 0.22	molar ratio, –, –	[F]	`mTORC1/Lac:Pyr` masking of `ULK1`; D OXPHOS efficiency
Energetic Stress	$\gamma_{\mathrm{pdh}},\ \gamma_{\mathrm{src}},\ \alpha_{\mathrm{gal}},\ \kappa_{\mathrm{glyc}},\ c_{\mathrm{akt}},\ K_{R,\mathrm{ATP}}$	0.72, 0.22, 0.74, 1.40, 0.08, 8.0	dimensionless	[F]	OXPHOS/glycolytic ATP & galactose shift [9]
ROS Damage Gate	$\beta_{\mathrm{LP}},\ \beta_h,\ \beta_{\mathrm{pH}},\ c_1,\ c_2,\ \theta_{\mathrm{GSH}},\ k_{\mathrm{dmg}}$	0.40, 0.88, 1.90, 0.24, 0.06, 1.38, 0.0012	–, h^{-1}	[F]	Threshold ROS conversion $W \to D$ above θ_{GSH} [11]
Clonal Division N	$\mu_{\mathrm{glyc}},\ \mu_{\mathrm{ox}},\ K_{\mathrm{ATP}},\ K_{W,\mathrm{div}},\ N_{\mathrm{cap}}$	0.062, 0.0185, 0.52, 0.56, 85.0	h^{-1}, –, fold	[M/A]	Lymphoblast doubling ($T_d \approx 11$–15 h) & cassette cap [1]
Supernatant R_{LP}	$k_{\mathrm{Warburg}},\ \alpha_h,\ k_{\mathrm{PDH}},\ \beta_{\mathrm{azd}},\ R_{\mathrm{media}}$	3.00, 0.46, 0.088, 2.80, 5.0	h^{-1}, –, ratio	[M/F]	Aerobic glycolysis vs. `AZD7545` `PDH` flux [10]
`FOXO1` Shuttle F_{nuc}	$k_{\mathrm{imp}},\ f_0,\ \alpha_{\mathrm{akt}},\ \phi_{\mathrm{SIRT1}},\ k_{\mathrm{exp}},\ \gamma_{\mathrm{acid}}$	0.056, 0.30, 0.30, 1.18, 0.032, 7.5	h^{-1}, –	[M/F]	`Akt` 14-3-3 export vs. `SIRT1` retention [3, 4]
Cristae SRC	$s_{\mathrm{basal}},\ k_{\mathrm{cristae}},\ \beta_{\mathrm{src}},\ k_{\mathrm{src,dec}},\ K_{\mathrm{SRC,mem}}$	0.044, 0.074, 1.32, 0.048, 0.68	h^{-1}, –	[M/F]	Cristae remodeling & $\mathrm{T_{SCM}}$ saturation [2]
Exhaustion E_{TOX}	$k_{\mathrm{exh}},\theta_{\mathrm{exh}},k_{\mathrm{warb,exh}},k_{\mathrm{acid,exh}},r_{\mathrm{epig},0},k_{\mathrm{rev,epig}}$	0.0038, 1.36, 0.0032, 0.22, 0.0032, 0.011	h^{-1}, –	[F]	Glycolytic/acidotic TOX^+ commitment vs. remodeling [5]
Perfusate pH / Flush	$k_{\mathrm{perf},0},k_{\mathrm{perf,fl}},k_{\mathrm{perf,LP}},k_{\mathrm{fl,LP}},\mathrm{pH}_{\mathrm{media}},k_{\mathrm{acid}},R_{\mathrm{buf}},N_{\mathrm{ref}}$	0.48, 1.80, 0.12, 0.85, 7.36, 0.007, 12.0, 48.0	h^{-1}, pH, fold	[A]	Hollow-fiber buffer exchange & bypass neutral flush

quotient law

$$\dot{h}(t) = -\lambda_{\mathrm{CART}}(t)\, h(t)\big(1 - h(t)\big),$$

$$\lambda_{\mathrm{CART}}(t) = \big(k_{\mathrm{mito},D}(1, R_{\mathrm{LP}}(t)) - k_{\mathrm{mito},W}\big) - (k_{\mathrm{rep,adv}} - 1)k_{\mathrm{bio},0}\Big(1 - \frac{M(t)}{K}\Big)_+. \tag{3}$$

In terms of the accumulated-selection coordinate $s(t) = \int_0^t \lambda_{\mathrm{CART}}(q)\, dq$, *intracellular depolarization evolves strictly as* $h(s) = [1 + ((1 - h_0)/h_0)e^s]^{-1}$, *crossing any target gate* $h_{\mathrm{gate}} \in (0, h_0)$ *at* $s^* = \ln[h_0(1 - h_{\mathrm{gate}})/(h_{\mathrm{gate}}(1 - h_0))]$.

The step-by-step derivation of Proposition 1 is provided in Appendix A.2. Inspection of (1) and the Stage-2 signaling equations (Appendix A.1) also clarifies why *co-actuation* of `PDK1` inhibition (g_{pdh}) and `SIRT1`/NAD^+ activation is required after switching: without `PDK1` inhibition ($u_{\mathrm{azd}} = 0 \implies g_{\mathrm{pdh}} = 0.15$), pyruvate remains blocked from entering the tricarboxylic acid cycle, collapsing both oxidative proliferation $\mu_{\mathrm{ox}} g_{\mathrm{pdh}} \Phi_{\mathrm{ox}}$ and NAD^+-coupled `FOXO1` nuclear retention; conversely, without `SIRT1` activation ($u_{\mathrm{sirt}} = 0$), intact mitochondrial biogenesis remains near basal $k_{\mathrm{bio},0}$ while residual depolarized organelles expand with replicative advantage $k_{\mathrm{rep,adv}} = 1.22$ via $\xi_D u_2$, causing $h(t)$ to rebound during Stage 2. All nominal parameters, initial conditions $\mathbf{x}(0) = [0.54, 0.42, 1.00, 22.0, 0.28, 1.30, 0.08, 7.32]^\top$ ($h_0 = 43.75\%$), and explicit provenance tags (**[M]** primary literature scale, **[F]** calibrated T-cell coupling, **[A]** operational cassette specification) are listed in Table 1.

3 Structural Bioinformatics Guiding Actuator Selection

A central design question for Stage-2 metabolic consolidation is why pharmacological actuation should target upstream metabolic kinases and deacetylases (`PDK1` and `SIRT1`/NAD^+) rather than attempting direct small-molecule pocket activation of the master stemness transcription factors `FOXO1` and `PGC-1`α. To ground our actuator parameterization in atomic coordinates, we audited experimental macromolecular crystallographic information files (`.cif`) from the Protein Data Bank alongside full-length AlphaFold Database v6 models and $N \times N$ Predicted Aligned Error (`PAE`) matrices [17, 18] (Figure 1, Table 2).

3.1 Druggable Allosteric Pockets in `PDK1` (`2Q8G`) and `SIRT1` (`4ZZJ`)

Coordinate inspection of human `PDK1` co-crystallized with the trifluoropropanamide inhibitor `AZD7545` (`PDB: 2Q8G`, Chain A, 368 resolved residues spanning `41`–`423` at 1.90 Å resolution [10]) confirms that `AZD7545` (`AZX`,

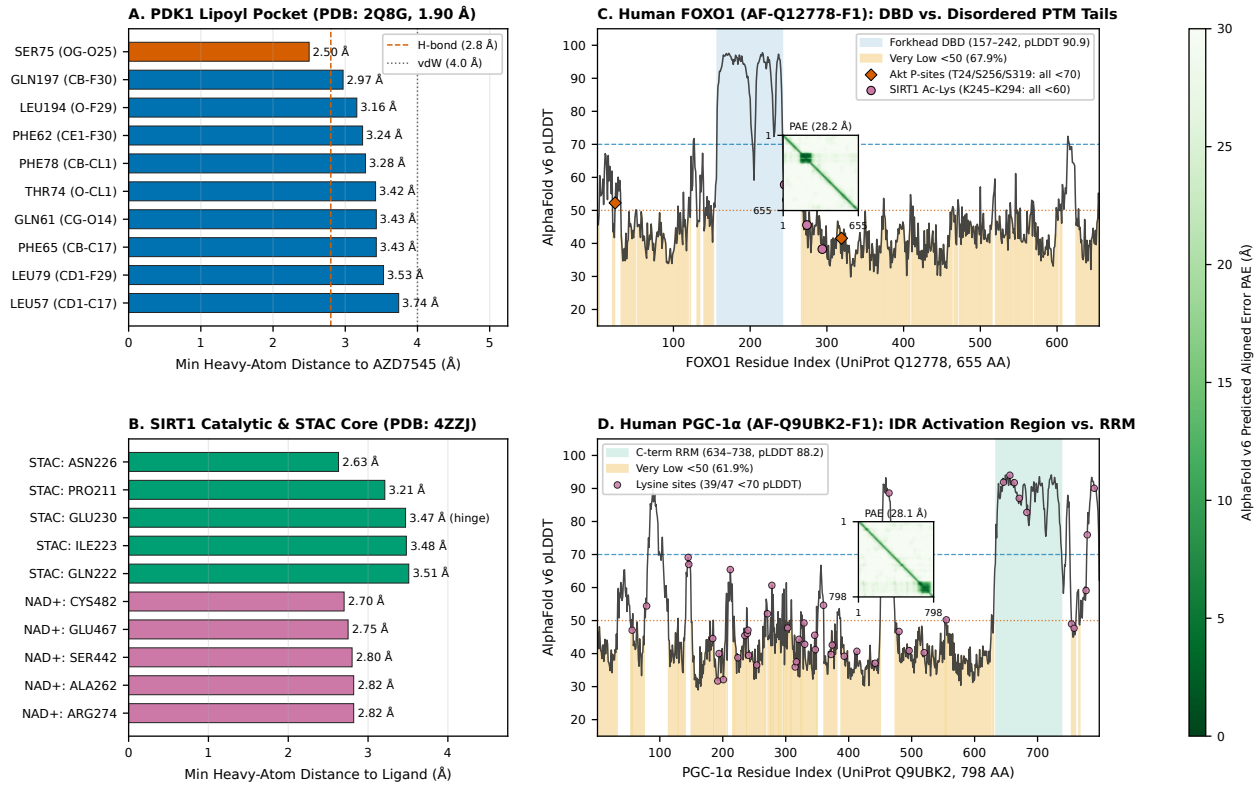


Figure 1: **Atomic Ligand-Contact and AlphaFold v6 pLDDT/PAE Structural Bioinformatics Analysis Guiding Stage-2 Actuator Selection.** **(A)** Heavy-atom contacts ($\leq$ 4.0 Å) between human PDK1 and the allosteric inhibitor AZD7545 (AZX) in PDB: 2Q8G (1.90 Å X-ray [10]), anchored by a 2.50 Å hydrogen bond at Ser75 (Oγ–O25) inside the N-terminal lipoyl-binding (L2) pocket. **(B)** Heavy-atom contacts ($\leq$ 4.0 Å) in human SIRT1 (PDB: 4ZZJ, 2.74 Å X-ray [15]) to the Sirtuin-Activating Compound 4TQ (Asn226 2.63 Å; allosteric hinge Glu230 3.47 Å) and active-site carba-NAD$^+$ (CNA). **(C)** Full-length human FOXO1 (AF-Q12778-F1-v6, 655 AA): only the Forkhead DNA-binding domain (residues 157–243, mean pLDDT = 90.62) is structured, whereas 9/9 Akt phosphorylation (Thr24, Ser256, Ser319) and SIRT1 deacetylation (Lys245–Lys294) sites reside in disordered segments (pLDDT $<$ 70; inset: 655 $\times$ 655 PAE matrix, off-diagonal mean 28.23 Å). **(D)** Full-length human PGC-1α (AF-Q9UBK2-F1-v6, 798 AA): only the C-terminal RNA-recognition motif (residues 634–738, mean pLDDT = 88.22) is structured, whereas 39/47 (82.98%) of lysine residues and the entire N-terminal/central activation region (1–633) are disordered (pLDDT $<$ 70).

31 heavy atoms) occupies the hydrophobic N-terminal lipoyl-binding (L2) pocket rather than the C-terminal ATP-binding cleft. Twelve residues form heavy-atom contacts $\leq$ 4.0 Å to AZD7545 (Figure 1A): a short 2.50 Å hydrogen bond between Ser75 (Oγ) and the ligand hydroxyl (O25), flanked by Gln197 (2.97 Å), Leu194 (3.16 Å), Phe62 (3.24 Å), Phe78 (3.28 Å), Thr74 (3.42 Å), Gln61 (3.43 Å), Phe65 (3.43 Å), Leu79 (3.53 Å), Leu57 (3.74 Å), Asp64 (3.92 Å), and Leu201 (3.92 Å). By severing PDK1 tethering to the dihydrolipoamide acetyltransferase (E2) core of the PDH complex, AZD7545 unblocks pyruvate oxidation without competing against millimolar intramitochondrial ATP.

Similarly, coordinate audit of the engineered human SIRT1 construct bound to Sirtuin-Activating Compound 4TQ and active-site carba-NAD$^+$ (PDB: 4ZZJ, Chain A, 344 resolved residues within 183–660 at 2.74 Å resolution [15]) resolves two distinct structured binding sites (Figure 1B): (i) the N-terminal three-helix STAC-

binding domain, where 4TQ (36 heavy atoms) contacts Asn226 (2.63 Å Nδ2–OAN hydrogen bond), Pro211 (3.21 Å), the critical electrostatic hinge residue Glu230 (3.47 Å Oε1–NAB), Ile223 (3.48 Å), and Gln222 (3.51 Å); and (ii) the Rossmann-fold NAD$^+$-binding pocket, where carba-NAD$^+$ (CNA, 44 heavy atoms) is coordinated by Cys482 (2.70 Å), Glu467 (2.75 Å), Ser442 (2.80 Å), Ala262 (2.82 Å), and Arg274 (2.82 Å).

Chemical-Probe Caveat for SIRT1 STACs: Early synthetic STACs such as SRT1720 were shown by Pacholec et al. [12] to exhibit fluorophore-dependent artifacts on TAMRA-labeled peptides and off-target pharmacology in biochemical panels, while Hubbard et al. [13] demonstrated that Glu230-mediated allosteric activation of native substrates requires bulky hydrophobic residues at positions +1/+6 C-terminal to the target acetyl-lysine. Consequently, in our model and postulated benchtop falsification protocol (§7), Stage-2 SIRT1 activation (u_{sirt}) primarily represents NAD$^+$ substrate repletion via nicotinamide riboside (NR, acting directly at the Cys482/Glu467/Arg274 Rossmann pocket), with synthetic STACs treated strictly as an optional tool-compound arm requiring orthogonal genetic (SIRT1 CRISPR knockout) and Glu230Lys point-mutant controls.

3.2 Domain Architecture and PTM-Tail Disorder of FOXO1 and PGC-1α

Full-length AlphaFold v6 structural audit reveals a sharp dichotomy between isolated nucleic-acid-binding domains and disordered post-translational modification (PTM) regions in both downstream effectors (Figure 1C–D, Table 2):

1. **Human FOXO1 (AF-Q12778-F1-v6, 655 AA):** Global mean pLDDT is 50.96, with 67.94% (445/655 AA) in the *Very Low* (< 50) band, 18.17% (119/655) in 50–70, and 13.89% (91/655) ≥ 70, accompanied by a high off-diagonal mean PAE = 28.23 Å (1.86% residue pairs < 5 Å). Contiguous domain segmentation (≥ 25 AA with pLDDT ≥ 70) identifies *one* structured domain: the winged-helix Forkhead DNA-binding domain (residues 157–243, 87 AA, mean pLDDT = 90.62, internal PAE = 4.70 Å). Crucially, **all 9/9 canonical regulatory PTM residues** controlling nuclear-cytoplasmic shuttling lie outside rigid tertiary pockets (pLDDT < 70): the three Akt 14-3-3 phosphosites (Thr24: 52.28, Ser256: 66.75, Ser319: 41.56) and all six CBP/p300/SIRT1 acetyl-lysines (Lys245: 57.78, Lys248: 53.19, Lys262: 55.19, Lys265: 52.41, Lys274: 45.56, Lys294: 38.28).
2. **Human PGC-1α (AF-Q9UBK2-F1-v6, 798 AA):** Global mean pLDDT is 52.74 (61.90% < 50, 17.17% in 50–70, 20.93% ≥ 70; off-diagonal mean PAE = 28.12 Å, 1.26% pairs < 5 Å). Segmentation detects *one* structured C-terminal RNA-recognition motif (RRM, residues 634–738, 105 AA, mean pLDDT = 88.22, internal PAE = 7.79 Å), whereas the entire 633-residue N-terminal and central transcriptional coactivation region—and 39/47 (82.98%) of all lysine residues across the protein (mean lysine pLDDT = 53.53)—is intrinsically disordered.

Rather than claiming that disordered transcription factors are universally "undruggable," this structural audit explains *mechanistically* why conventional pocket-based small-molecule discovery lacks preformed allosteric cavities in the PTM control regions of FOXO1 and PGC-1α, whereas the structured catalytic cleft of SIRT1 (PDB: 4ZZJ) is evolutionarily specialized to capture flexible acetyl-lysine peptide motifs.

Table 2: **Verified Atomic-Coordinate and AlphaFold v6 Structural Bioinformatics Summary.** All metrics are computed directly from local `mmCIF` and `PAE` JSON coordinate files by `audit_all_structures()` in `simulator/generate_paper2_verified_benchmarks.py`.

Target Protein	Structure / Accession	Length / Res.	Structured Domain(s)	Disorder / PAE Metrics	Verified Key Contacts or PTM Sites
PDK1 (Kinase)	PDB: 2Q8G (X-ray)	368 AA (1.90 Å)	Lipoyl (L2) + Kinase core	Rigid co-crystal (41–423)	AZD7545: Ser75 (2.50 Å), Gln197 (2.97 Å), Leu194 (3.16 Å)
SIRT1 (Deacetylase)	PDB: 4ZZJ (X-ray)	344 AA (2.74 Å)	STAC 3-helix + Rossmann	Rigid co-crystal (183–660)	4TQ: Asn226 (2.63 Å), Glu230 (3.47 Å); CNA: Cys482 (2.70 Å)
FOXO1 (TF)	AF-Q12778-F1-v6	655 AA (AF v6)	Forkhead DBD 157–243 (90.6)	67.9% < 50; PAE = 28.23 Å	9/9 Akt (T24/S256/S319) & Ac-Lys (K245–K294) < 70 pLDDT
PGC-1α (Coact.)	AF-Q9UBK2-F1-v6	798 AA (AF v6)	C-term RRM 634–738 (88.2)	61.9% < 50; PAE = 28.12 Å	1–633 activation IDR; 39/47 (83.0%) lysines < 70 pLDDT

Table 3: **Nominal 168-Hour *In Silico* Protocol Sweep, Target Ablations, and Discrete-Adjoint PMP Benchmark.** Release specification: $N \geq 50\times$, $\mathrm{T_{SCM}} \geq 68\%$, $\mathrm{TOX^+} \leq 15\%$, SRC $\geq 3.5\times$, and $h \leq 10\%$.

Simulated Regimen	t_{gate} (h)	$s(t_{\mathrm{gate}})$	**Fold** $N(168\,\mathrm{h})$	$\mathrm{T_{SCM}}$ (%)	$\mathrm{TOX^+}$ (%)	SRC (×)	$h(168\,\mathrm{h})$ (%)	**5-Spec Pass**
Unphased Warburg ($u_1 = 0, u_2 = 0$)	–	–	81.77×	3.67%	76.13%	0.69×	66.76%	Fail (4/5)
Premature Oxidative ($u_1 = 0, u_2 = 1$ from 0 h)	0.00	0.0000	9.58×	43.92%	27.64%	1.22×	64.81%	Fail (5/5)
Continuous Urolithin A Monotherapy ($u_1 = 1, u_2 = 0$)	–	–	80.60×	31.10%	41.03%	2.26×	0.04%	Fail (3/5)
Fixed Early Switch ($t_{\mathrm{sw}} = 36$ h)	36.12	1.4696	46.11×	72.02%	9.32%	3.58×	23.51%	Fail (N, h)
Fixed Intermediate Switch ($t_{\mathrm{sw}} = 48$ h)	48.05	2.0009	54.72×	73.17%	10.16%	4.23×	14.87%	Fail ($h > 10\%$)
Closed-Loop Bioreactor-BCR ($h \leq 4.5\%$)	**66.02**	**2.8248**	**62.80×**	**73.20%**	**11.75%**	**4.87×**	**6.73%**	**PASS** (5/5)
Fixed Late Switch ($t_{\mathrm{sw}} = 96$ h)	96.10	4.2392	70.77×	70.26%	15.77%	5.13×	1.54%	Fail ($\mathrm{TOX^+}$)
Fixed Very Late Switch ($t_{\mathrm{sw}} = 120$ h)	120.12	5.3857	75.04×	65.03%	21.08%	4.78×	0.46%	Fail ($\mathrm{T_{SCM}}$, $\mathrm{TOX^+}$)
Ablate Stage-2 AZD7545 (PDB: 2Q8G, $u_{\mathrm{azd}} = 0$)	66.02	2.8248	37.47×	30.72%	21.25%	2.66×	10.82%	Fail (5/5)
Ablate Stage-2 SIRT1/NR (PDB: 4ZZJ, $u_{\mathrm{sirt}} = 0$)	66.02	2.8248	52.25×	17.45%	28.46%	1.62×	36.27%	Fail (4/5)
Discrete-Adjoint PMP Optimal (200 vars, $\alpha = 0.58$)	Cont.	–	**71.15×**	**74.46%**	**14.08%**	**5.33×**	**0.09%**	**PASS** (5/5)

4 In Silico Protocol Sweep, Target Ablations, and Adjoint Pareto Optimality

4.1 Nominal Switching Sweep and Mechanistic Component Ablations

To evaluate whether two-stage metabolic gating resolves the simulated proliferation–exhaustion conflict, we integrated the nominal 8-state system (RK4, $N = 1{,}000$ steps, $\Delta t = 0.168$ h) across 14 open-loop, closed-loop, and ablated regimens against five simultaneous manufacturing release criteria: viable expansion $N_{\mathrm{cells}}(168\,\mathrm{h}) \geq 50.0\times$, stemness $\mathrm{T_{SCM}}(168\,\mathrm{h}) \geq 68.0\%$, exhaustion $E_{\mathrm{TOX}}(168\,\mathrm{h}) \leq 15.0\%$, SRC $\geq 3.5\times$, and depolarized organelle fraction $h(168\,\mathrm{h}) \leq 10.0\%$ (Figure 2A–B, Table 3).

Unphased Warburg expansion ($u_1 = u_2 = 0$) achieves maximal cell yield (81.77×) but drives severe mitochondrial depolarization ($h = 66.76\%$), supernatant lactatemia ($R_{\mathrm{LP}} = 30.02$, pH = 6.92), nuclear FOXO1 exclusion ($F_{\mathrm{nuc}} = 30.48\%$), $\mathrm{TOX^+}$ commitment (76.13%), and $\mathrm{T_{SCM}}$ collapse (3.67%). Conversely, premature oxidative switching at Hour 0 ($u_1 = 0, u_2 = 1$) stalls clonal division at 9.58× (< 50× floor) while failing to clear baseline depolarized mitochondria ($h = 64.81\%$). Continuous urolithin A monotherapy ($u_1 = 1, u_2 = 0$, without Stage-2 galactose/AZD7545/SIRT1 consolidation) thoroughly purges depolarized mitochondria ($h = 0.04\%$) and preserves blastogenesis (80.60×), but because high glycolytic flux continues to drive Warburg exhaustion (41.03%) and prevents full cristae SRC maturation (2.26×), simulated $\mathrm{T_{SCM}}$ plateaus at 31.10%.

Under nominal closed-loop gating ($h(t) \leq 0.045$, $N_{\mathrm{cells}}(t) \geq 14.0\times$, $\mathrm{pH}(t) \geq 7.10$), **Bioreactor-BCR** transitions from Media-A to Media-B at $t_{\mathrm{gate}} = 66.02$ h (accumulated selection integral $s(t_{\mathrm{gate}}) = 2.8248$), achieving 62.80× fold expansion, 73.20% $\mathrm{T_{SCM}}$, SRC = 4.87×, $E_{\mathrm{TOX}} = 11.75\%$, and $h(168\,\mathrm{h}) = 6.73\%$—satisfying all five release criteria simultaneously. Ablating Stage-2 AZD7545 ($u_{\mathrm{azd}} = 0$ after 66.02 h) leaves PDH phosphorylated ($g_{\mathrm{pdh}} = 0.15$), cutting expansion in galactose-shifted media to 37.47× and $\mathrm{T_{SCM}}$ to 30.72%. Ablating Stage-2 SIRT1/NR activation ($u_{\mathrm{sirt}} = 0$ after 66.02 h) allows D to out-replicate W ($k_{\mathrm{rep,adv}} = 1.22$), causing h to rebound from 4.50% at the gate to 36.27% at 168 h and collapsing $\mathrm{T_{SCM}}$ to 17.45%.

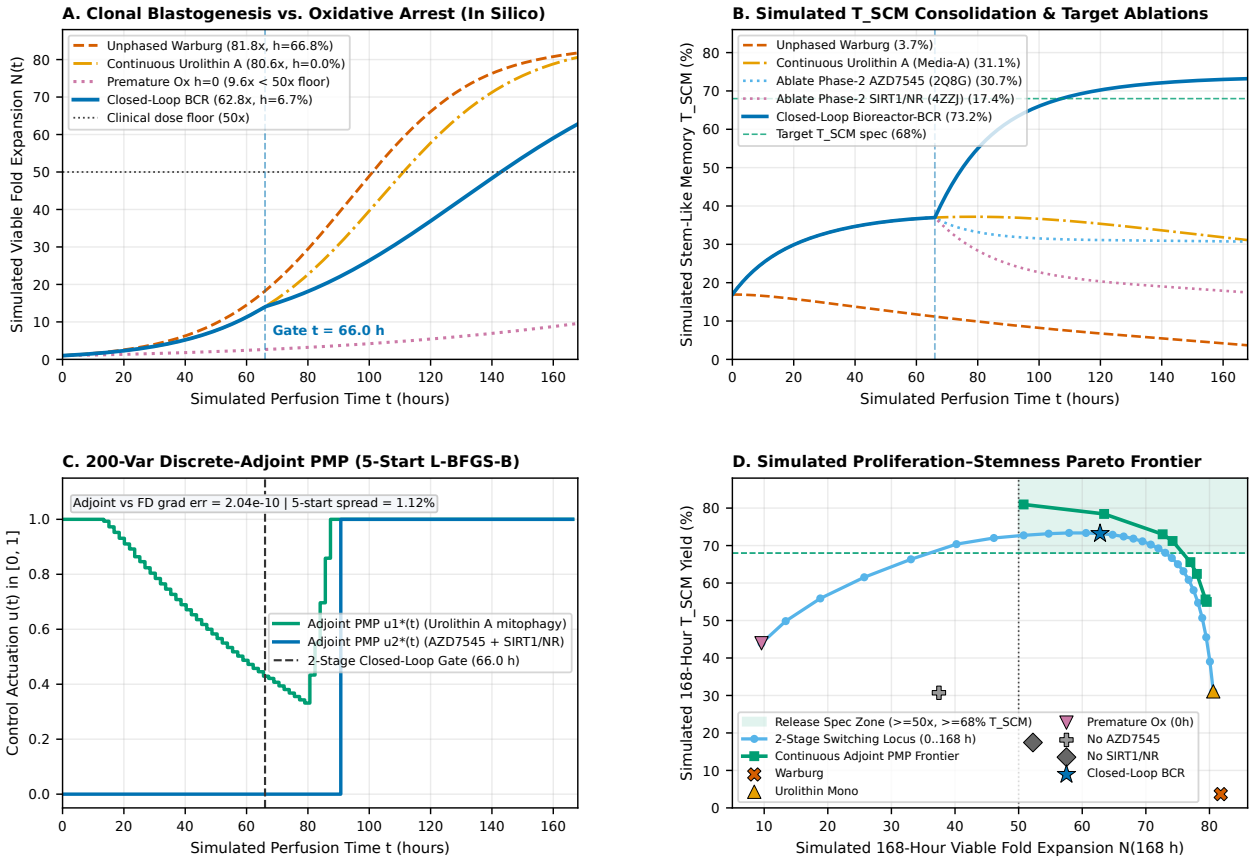


Figure 2: ***In Silico* 168-Hour Perfusion Trajectories, Component Ablations, and 200-Variable Discrete-Adjoint Pareto Frontier.** **(A)** Simulated clonal fold expansion $N_{\text{cells}}(t)$ showing Warburg glycolytic expansion (81.77×, $h = 66.76\%$), premature Day-0 oxidative forcing (9.58× < 50× clinical floor), continuous urolithin A monotherapy (80.60×), and closed-loop **Bioreactor-BCR** switching at $t_{\text{gate}} = 66.02$ h (62.80×, $h = 6.73\%$). **(B)** Simulated $\text{T}_{\text{SCM}}(t)$ consolidation comparing closed-loop gating (73.20%) against continuous urolithin A without Stage-2 oxidative consolidation (31.10%) and single-target Stage-2 ablations of `AZD7545` (30.72%) or `SIRT1/NR` (17.45%). **(C)** Unconstrained 200-variable discrete-adjoint Pontryagin optimal control schedule $(u_1^*(t), u_2^*(t)$ across 100 intervals; maximum adjoint vs. central finite-difference gradient error across all 200 variables 2.04×10^{-10}, 5-start `L-BFGS-B` objective spread 1.12%). **(D)** Simulated proliferation–stemness Pareto frontier comparing continuous PMP modulation against the 2-stage switching locus ($t_{\text{sw}} \in [0, 168$ h$]$) and discrete baseline/ablation protocols.

4.2 200-Variable Discrete-Adjoint Optimal Control and Pareto Frontier

To test whether a simple two-stage pinch-valve switch sacrifices substantial performance relative to an unconstrained, continuously modulated multi-pump schedule $(u_1(t), u_2(t)) \in [0, 1]^2$, we discretized $[0, 168$ h$]$ into $N_{\text{ocp}} = 100$ piecewise-constant control intervals (200 decision variables) and minimized the scalarized

terminal-plus-effort Bolza objective implemented in `solve_discrete_adjoint_pareto()`:

$$J_\alpha(\mathbf{u}) = -\alpha\frac{\mathrm{T_{SCM}}(\mathbf{x}(T))}{100} - (1-\alpha)\frac{N_{\mathrm{cells}}(\mathbf{x}(T))}{75.0} + 0.45\Big(\frac{h(T)}{0.10}\Big)^2 + 0.35\,E_{\mathrm{TOX}}(T)^2$$
$$+ 2.50\Big(\frac{\max_\epsilon(0,\ 50.0 - N_{\mathrm{cells}}(T))}{50.0}\Big)^2 + \frac{0.008}{N_{\mathrm{ocp}}}\sum_{k=0}^{N_{\mathrm{ocp}}-1}\big(u_{1,k}^2 + u_{2,k}^2\big), \tag{4}$$

with smooth hinge regularization $\max_\epsilon(0, z)$ and exact stage-by-stage `RK4` discrete-adjoint sensitivity recursion detailed in Appendix A.3. Verifying the discrete-adjoint gradient against central finite differences ($\epsilon_{\mathrm{FD}} = 10^{-6}$) across *all* 200 *decision variables* yields a maximum discrepancy of 2.04×10^{-10}. Launching `L-BFGS-B` at the balanced Pareto weight $\alpha = 0.58$ from 5 diverse initializations (two-stage step at 66.0 h, uniform interior $u_1 = u_2 = 0.5$, early step at 36.0 h, late step at 96.0 h, and pseudo-random interior schedule $\mathcal{U}[0.1, 0.9]$) converges to a tight objective spread of 1.12% ($J^*_{0.58} = -0.81366$ vs. two-stage closed-loop $J_{0.58} = -0.58290$, Table 3).

Notably, the Pontryagin-optimal control profile (Figure 2C) independently discovers a *two-phase temporal structure*: it holds $u_2^*(t) = 0$ throughout Hours 0–90.7 while modulating $u_1^*(t)$ along a smooth interior arc ($1.0 \to 0.33$, avoiding the small c_{akt} glycolytic proliferation penalty once $h(t)$ is low), executes a brief 15-hour co-actuation overlap ($u_1^* = 1, u_2^* = 0 \to 1$) to block $k_{\mathrm{rep,adv}}$ rebound during initial mitochondrial biogenesis, and holds $u_1^* = u_2^* = 1$ to Hour 168. Sweeping $\alpha \in [0.08, 0.92]$ (Figure 2D) shows that while continuous multi-pump modulation expands yield from 62.80× to 71.15× (+1.26% $\mathrm{T_{SCM}}$), the single-valve two-stage locus already enters well inside the 5-criteria clinical release zone without requiring continuous microfluidic drug mixing.

5 Heterogeneous Oncology Cohort, Ratiometric `ORR` Observability, and Adaptive Extended-Horizon Rescue

5.1 Volumetric Respirometry Bias vs. Cell-Density Cancellation in Ratiometric `ORR`

In a physical closed-cassette perfusion bioreactor, inline dissolved-O_2 optodes do *not* measure per-cell specific respiration directly; they measure *volumetric* oxygen depletion proportional to total viable cell count N_{cells} times per-cell oxidative flux, while inline near-infrared (`NIR`) backscatter or capacitance probes estimate cell concentration subject to donor-specific lymphoblast-size calibration bias α_{OD}. To evaluate observability under realistic bioprocess physics and statistical power, we scaled our virtual cohort to $N = 1{,}000$ oncology apheresis donors ($h_0 \sim \mathcal{U}[0.20, 0.62]$, $M_0 \sim \mathcal{U}[0.85, 1.12]$, $R_{\mathrm{LP},0} \sim \mathcal{U}[18, 28]$, $F_{\mathrm{nuc},0} \sim \mathcal{U}[0.22, 0.36]$, $S_{\mathrm{SRC},0} \sim \mathcal{U}[1.05, 1.50]$, $E_{\mathrm{TOX},0} \sim \mathcal{U}[0.05, 0.14]$) with unmodeled kinetic perturbations ($\mu_{\mathrm{glyc}} \times \mathcal{U}[0.66, 1.36]$, $\mu_{\mathrm{ox}} \times \mathcal{U}[0.84, 1.16]$, $k_{\mathrm{mito},D,\max} \times \mathcal{U}[0.68, 1.36]$, $k_{\mathrm{Warburg}} \times \mathcal{U}[0.82, 1.22]$, $k_{\mathrm{warb,exh}} \times \mathcal{U}[0.82, 1.22]$) and unmodeled ±6% optical-density scattering bias $\alpha_{\mathrm{OD}} \sim \mathcal{U}[0.94, 1.06]$. The Huber-EKF runs strictly on `NOMINAL_PARAMS` ($\hat{\alpha}_{\mathrm{OD}} = 1.00$) and never observes any donor's private parameters or initial state.

Sampled every $\Delta t_{\mathrm{obs}} = 0.30$ h (18 min) under correlated AR(1) noise ($\rho = 0.65$; exact measurement equations and heteroscedastic noise models in Appendix A.4), the physical 4-channel cassette suite observes amperometric supernatant $y_1 = R_{\mathrm{LP}} + \nu_1$, volumetric optode respiration $y_2 = N_{\mathrm{cells}}\,\mathrm{ATP_{ox}}(\mathbf{x}, u_2) + \nu_2$, biased NIR optical density $y_3 = \alpha_{\mathrm{OD}} N_{\mathrm{cells}} + \nu_3$, and phenol-red optical transmission $y_4 = \mathrm{pH} + \nu_4$. Dividing volumetric respiration y_2 by biased optical density y_3 yields $y_2/y_3 \approx \alpha_{\mathrm{OD}}^{-1}(W + \eta_D D)(1 + \gamma_{\mathrm{pdh}} u_2)(1 + \gamma_{\mathrm{src}}(S_{\mathrm{SRC}} - 1)_+)$, which couples only to the sum $W + \eta_D D$ and inherits the permanent $\alpha_{\mathrm{OD}}^{-1}$ lymphoblast-size bias.

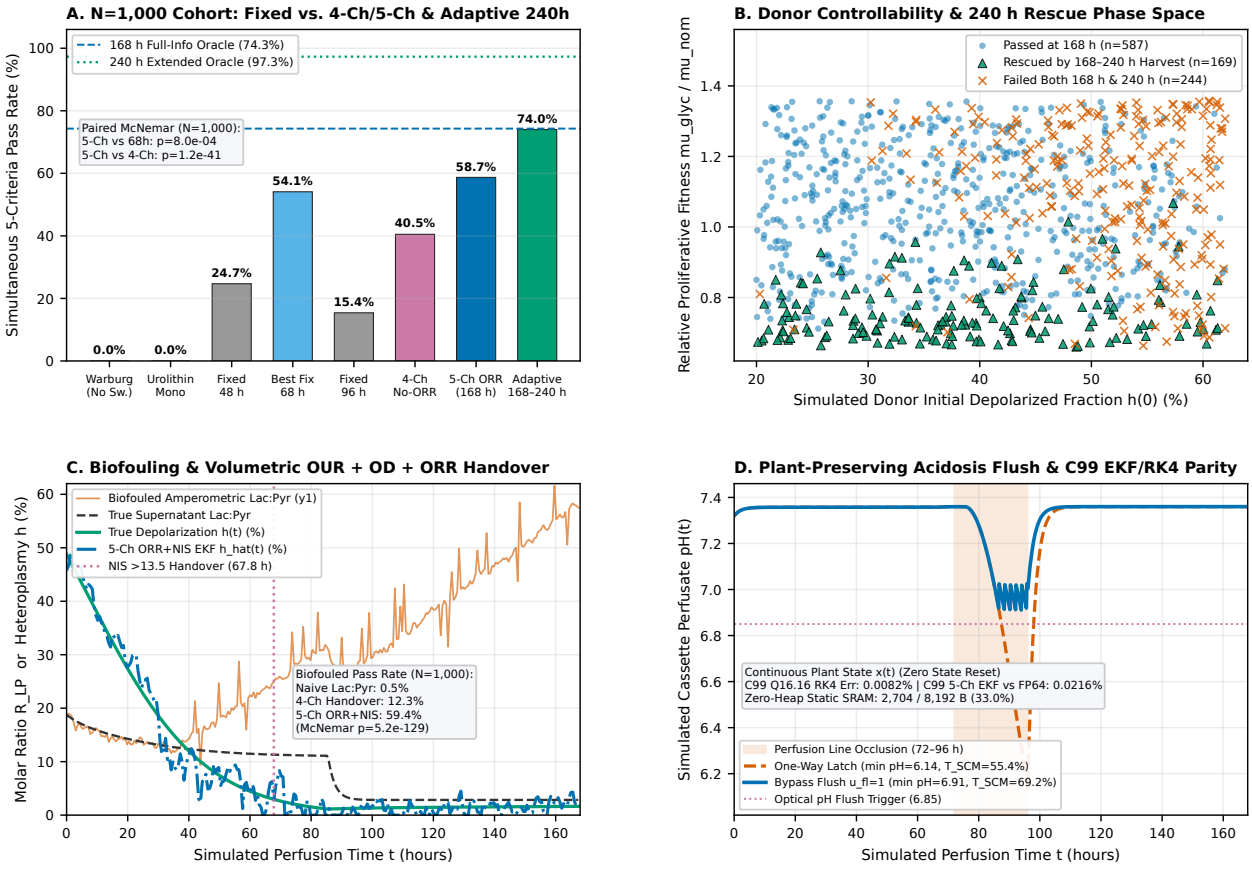


Figure 3: $N = 1{,}000$ **Heterogeneous Oncology Donor Cohort, Physical Volumetric `OUR/OD/ORR` Observability Under Biofouling, and Extended-Horizon Rescue.** **(A)** Simultaneous 5-criteria manufacturing pass rates across $N = 1{,}000$ simulated oncology apheresis donors comparing unphased baselines (0.0%), open-loop fixed timers (peaking at 54.1% at 68 h), physical 4-channel EKF without `ORR` (40.5%), 5-channel `ORR`-augmented Huber-EKF at 168 h (58.7%; exact paired McNemar $p = 8.0 \times 10^{-4}$ vs. 68 h, $p = 1.2 \times 10^{-41}$ vs. 4-channel), and `OD`-triggered adaptive harvest (168–240 h, 74.0%; 168 h oracle 74.3%, 240 h oracle 97.3%). **(B)** 2D oncology donor controllability phase space $(h_0, \mu_{\text{glyc}}/\mu_{\text{nom}})$ partitioning the 1,000 donors into 168 h passers ($n = 587$), hypoproliferative donors rescued by 168–240 h adaptive harvest ($n = 169$), and unrescued severe-burden donors ($n = 244$). **(C)** Representative biofouling trajectory where enzymatic `Lac:Pyr` drift triggers NIS > 13.5 residual handover at 67.8 h to the optical quartet (OUR_{vol}`+OD+pH+ORR`), preserving a 59.4% cohort pass rate ($p = 5.2 \times 10^{-129}$ vs. 12.3% 4-channel handover). **(D)** Plant-preserving bypass neutral flush ($u_{\text{fl}} = 1$, zero state reset) during perfusion-line occlusion (72–96 h), lifting min pH from 6.14 to 6.91, alongside freestanding C99 `Q16.16 RK4` (0.0082%) and 5-channel EKF (0.0216%) parity.

Augmenting the cassette with a fifth optical channel—quartz-window two-band FAD/(FAD + NAD(P)H) autofluorescence [14]—resolves this structural collinearity. Because bulk suspension fluorescence in both spectral bands scales linearly with illuminated cell concentration ($I_{\text{FAD}} = \beta_{\text{opt}} N_{\text{cells}} W$ and $I_{\text{NAD(P)H}} = \beta_{\text{opt}} N_{\text{cells}} [D + c_{\text{glyc}} g_{\text{glc}}(u_2)]$), the unknown proportionality factor $\beta_{\text{opt}} N_{\text{cells}}$ *cancels identically* in the ratio:

$$y_5 = \frac{I_{\text{FAD}}}{I_{\text{FAD}} + I_{\text{NAD(P)H}}} + \nu_5 = \frac{W}{W + D + c_{\text{glyc}}\, g_{\text{glc}}(u_2)} + \nu_5, \tag{5}$$

with $\sigma_5 = 0.012$, nominal $c_{\text{glyc}} = 0.35$, and $\pm 8\%$ unmodeled donor variation in c_{glyc}. Across the six core gating

states $[W, D, N_{\text{cells}}, R_{\text{LP}}, S_{\text{SRC}}, \text{pH}]$, the 64-hour empirical Observability Gramian quantifies this mechanism: without ORR, $\kappa(\mathcal{G}_{4\text{ch}}) = 5.37 \times 10^5$ and degrades 13.4× upon R_{LP} electrode fouling to $\kappa(\mathcal{G}_{3\text{ch,foul}}) = 7.21 \times 10^6$ ($\sigma_{\min} = 1.34 \times 10^{-5}$), yielding $\hat{h}$ RMSE = 6.48% (unfouled) and 16.18% (biofouled). Adding ratiometric ORR (5) supplies an orthogonal sensitivity row independent of N_{cells} and α_{OD}, improving post-fouling Gramian conditioning 8.55× ($\kappa = 8.43 \times 10^5$) and holding $\hat{h}$ RMSE to 2.08% (unfouled) and 2.15% (biofouled).

5.2 Explicit Switching Law, Guard Ablation, and $N = 1{,}000$ Paired McNemar Benchmarks

To ensure complete controller transparency between Python and C99, the closed-loop Stage-1 → 2 transition requires 4 consecutive frames (1.2 h persistence quorum) with $y_4 \geq 7.10$ and satisfaction of either the primary reachability gate or an explicit late-glycolysis exhaustion guard:

$$C_{\text{switch}}(t) = \left(\hat{h} \leq 0.056 \ \wedge \ \hat{N}_{\text{cells}} \geq N_{\text{req}}(t, \hat{\mu}_{\text{rel}})\right) \ \vee \ \underbrace{\left(t \geq 83 \text{ h} \ \wedge \ \hat{h} \leq 0.072 \ \wedge \ \hat{N}_{\text{cells}} \geq N_{\text{guard}}\right)}_{\text{Late-glycolysis TOX}^+ \text{ ceiling guard}}, \quad (6)$$

where $\hat{\mu}_{\text{rel}}(t)$ is the online-estimated donor division rate relative to nominal and $N_{\text{req}}(t, \hat{\mu}_{\text{rel}})$ is the minimum fold expansion at hour t required to reach $N_{\text{target}} = 52.5\times$ (buffering ±6% OD bias above the true 50× floor; exact closed-form expressions in Appendix A.4).

Across all $N = 1{,}000$ oncology donors (Figure 3A, Table 4), unphased Warburg, premature Hour-0 oxidative switching, and continuous urolithin A monotherapy achieve 0.0% (0/1,000) 5-criteria pass rates. Among nine open-loop fixed timers (24–120 h), performance peaks sharply at $t_{\text{sw}} = 68$ h (54.1%, 541/1,000) and 76 h (52.8%). Because Warburg-flux and OD biases corrupt $\hat{h}$ without ORR, the 4-channel Huber-EKF achieves only 40.5% (405/1,000). The 5-channel ORR Huber-EKF raises the 168-hour pass rate to 58.7% (587/1,000), significantly outperforming both the retrospective-best 68 h fixed timer (+4.6%, paired 95% CI [+1.97%, +7.23%], exact two-sided McNemar $p = 8.04 \times 10^{-4}$; 114 discordant wins vs. 68 losses) and the 4-channel No-ORR EKF (+18.2%, 95% CI [+15.58%, +20.82%], $p = 1.19 \times 10^{-41}$; 197 vs. 15). Ablating the $t \geq 83.0$ h exhaustion guard in (6) drops the 5-channel pass rate from 58.7% to 49.9% (−8.8%, 95% CI [−10.58%, −7.02%], $p = 1.47 \times 10^{-25}$), confirming that capping glycolytic dwell time is essential to prevent TOX$^+$ > 15% in high-h_0 donors. Under 100% amperometric Lac:Pyr biofouling ($t_{\text{foul}} \sim \mathcal{U}[30, 48]$ h), naive Lac:Pyr gating collapses to 0.5% (5/1,000) and 4-channel NIS handover yields 12.3% (123/1,000), whereas 5-channel ORR+NIS handover ($\text{NIS}_1 > 13.5$ for 3 frames, 1,000/1,000 handovers) preserves 59.4% (594/1,000) (+47.1% vs. No-ORR handover, $p = 5.22 \times 10^{-129}$; +5.3% vs. unfouled 68 h fixed timer, $p = 1.81 \times 10^{-4}$).

5.3 Direct Test of Extended-Horizon (168–240 h) Rescue for Hypoproliferative Donors

Evaluating the full-information donor oracle at 168 h shows that 74.3% (743/1,000) of the cohort is controllable within 7 days, leaving 25.7% (257/1,000) infeasible at fixed 168 h (median $h_0 = 51.19\%$, $\mu_{\text{glyc}}/\mu_{\text{nom}} = 0.773\times$ vs. 38.79% and 1.059× in controllable donors). Rather than leaving extended culture as an untested conjecture, we simulated an adaptive variable-harvest policy ($T_{\text{plan}} = 240$ h, $N_{\text{target}} = 53.5\times$, $N_{\text{guard}} = 14.0\times$) that harvests at 168 h whenever $\hat{N}_{\text{cells}}(168 \text{ h}) \geq 53.5\times$ with ≥ 36 h Stage-2 consolidation, and otherwise continues Stage-2 perfusion up to 240 h until $\hat{N}_{\text{cells}}(t) \geq 53.5\times$. Extending the horizon raises the full-information oracle ceiling from 74.3% to 97.3% (973/1,000)—proving that 89.5% (230/257) of 168 h-infeasible runs fail solely from division-time truncation, while only 2.7% (27/1,000; median $h_0 = 57.14\%$, fast Warburg exhaustion $\mu_{\text{glyc}}/\mu_{\text{nom}} = 1.189\times$) are intrinsically refractory even at 240 h. In closed loop, OD-guided adaptive harvest extends culture past 168 h in only 22.5% of donors yet rescues 169 previously failing hypoproliferative runs

Table 4: **Simulated $N = 1{,}000$ Heterogeneous Oncology Apheresis Cohort, Guard/Sensor Ablations, Biofouling Handover, and Extended-Horizon (168–240 h) Rescue.** Medians and interquartile ranges (IQR) across all 1,000 donors (seed `20261111`).

Cohort Control Policy ($N = 1{,}000$ **Donors**)	**Pass Rate**	**Fold** N **Med [IQR]**	T_{SCM} **Med [IQR]**	TOX^+ **Med**	h_{end} **Med**	$\hat{h}$ **RMSE**	**Gate** t_{sw} **Med [IQR]**
Unphased Warburg ($u_1 = 0, u_2 = 0$, 168 h)	0.0% (0/1000)	80.7× [74.6, 83.7]	4.1% [1.8, 6.7]	75.0%	63.7%	–	–
Continuous Urolithin A Monotherapy (168 h)	0.0% (0/1000)	81.3× [72.2, 83.9]	30.5% [28.9, 32.1]	42.0%	0.03%	–	–
Open-Loop Fixed Timer (48 h)	24.7% (247/1000)	55.7× [42.1, 66.1]	72.9% [71.1, 74.2]	10.9%	13.1%	–	48.0 h (fixed)
Open-Loop Fixed Timer (60 h)	45.0% (450/1000)	61.8× [47.2, 71.6]	72.9% [71.5, 73.9]	11.9%	7.5%	–	60.0 h (fixed)
Best Open-Loop Fixed Timer (68 h)	**54.1%** (541/1000)	64.9× [50.3, 74.3]	72.5% [71.3, 73.6]	12.7%	5.1%	–	68.0 h (fixed)
Open-Loop Fixed Timer (84 h)	42.2% (422/1000)	69.5× [55.0, 77.5]	71.2% [70.0, 72.2]	14.6%	2.3%	–	84.0 h (fixed)
Open-Loop Fixed Timer (96 h)	15.4% (154/1000)	72.1× [58.1, 79.2]	69.5% [68.3, 70.7]	16.5%	1.2%	–	96.0 h (fixed)
Ablation: Physical 4-Ch EKF (**No** `ORR`, 168 h)	40.5% (405/1000)	68.9× [58.4, 75.8]	70.7% [66.2, 72.4]	14.9%	1.81%	6.48%	84.6 h [66.5, 100.7]
Ablation: 5-Ch `ORR` EKF (**No** 83 **h Guard**, 168 h)	49.9% (499/1000)	66.5× [58.4, 73.1]	71.1% [68.3, 72.7]	14.3%	3.81%	2.12%	79.4 h [63.0, 102.3]
Closed-Loop 5-Ch `ORR` **Huber-EKF (168 h)**	**58.7%** (587/1000)†	66.4× [56.2, 73.1]	71.7% [70.2, 72.8]	13.7%	4.34%	**2.08%**	79.4 h [63.0, 90.6]
Biofouled `Lac:Pyr`: Naive Threshold Gate	0.5% (5/1000)	81.2× [72.2, 83.9]	30.5% [29.0, 32.1]	42.0%	0.03%	47.26%	50.7 h ($n = 5$)
Biofouled `Lac:Pyr`: 4-Ch No-`ORR` NIS Handover	12.3% (123/1000)	74.2× [62.6, 80.8]	65.2% [60.3, 69.0]	20.9%	0.57%	16.18%	115.8 h [99.9, 129.9]
Biofouled `Lac:Pyr`**: 5-Ch** `ORR` **+ NIS Handover**	**59.4%** (594/1000)‡	67.0× [56.0, 73.5]	71.6% [70.2, 72.8]	13.9%	3.99%	**2.15%**	82.1 h [64.5, 90.6]
Adaptive Horizon (168–240 h) 5-Ch `ORR` **EKF**	**74.0%** (740/1000)§	65.7× [55.3, 73.1]	72.4% [71.2, 73.4]	13.1%	4.87%	**2.06%**	73.8 h [62.0, 84.6]
Full-Information Donor Oracle Ceilings (168 h / 240 h)	74.3% (743/1000) at fixed 168 h; 97.3% (973/1000) with 168–240 h extended harvest (2.7% intrinsically refractory)						

Exact two-sided paired McNemar tests: † $p = 8.04 \times 10^{-4}$ vs. 68 h fixed (+4.6%), $p = 1.19 \times 10^{-41}$ vs. 4-Ch No-`ORR` (+18.2%); ‡ $p = 5.22 \times 10^{-129}$ vs. 4-Ch handover; § $p = 2.08 \times 10^{-33}$ vs. 168 h EKF.

(green triangles in Figure 3B), lifting the closed-loop pass rate from 58.7% to 74.0% (740/1,000) (+15.3%, 95% CI [+12.81%, +17.79%], exact McNemar $p = 2.08 \times 10^{-33}$).

6 Plant-Preserving Acidosis Rescue and Deterministic Fixed-Point Firmware

6.1 Continuous-Plant Neutral Bypass Flush Under Perfusion Occlusion

As established in Paper I [11], restoring a microcontroller's internal estimator memory after a fault cannot reset biological state variables; actuator recovery must operate strictly through continuous plant dynamics $\mathbf{x}(t)$. In hollow-fiber CAR-T cassettes, a primary post-switch hazard is partial peristaltic line occlusion or gas-transfer fouling during high-density Stage-2 expansion ($N_{cells} > 20\times$). We simulated a 24-hour Media-B perfusion-line occlusion across Hours 72–96 (92% drop in main-line exchange $k_{perf,0}$ and $k_{perf,LP}$, loss of `AZD7545` delivery $u_{azd,eff} = 0$, and hypoxic glycolytic surge $1.65 \times k_{acid}$, Figure 3D).

Under a one-way valve latch ($u_{fl} = 0$), cassette perfusate pH(t) plunges to min pH = 6.14, triggering severe acidotic `FOXO1` nuclear export and TOX^+ commitment (Appendix A.1, $E_{TOX} = 32.28\%$, $T_{SCM} = 55.39\%$). When the supervisory controller monitors optical phenol-red pH(t) ≤ 6.85 and opens the auxiliary neutral-buffer bypass flush ($u_{fl} = 1$, reverting to protective Stage-1 media for a 6-hour stabilization window without altering $\mathbf{x}(t)$ discontinuously), min pH is arrested at 6.91, limiting E_{TOX} to 16.11% and rescuing simulated T_{SCM} to 69.16% ($N_{cells} = 70.59\times$, $h = 8.14\%$).

6.2 Freestanding FPU-Free C99 `RK4` and 5-Channel Huber-EKF Verification

Unlike implantable neural devices or solar agricultural sensors, a benchtop GMP cell-therapy bioreactor is wall-powered; embedded microwatt sleep budgets are not a bioprocess bottleneck. Instead, regulatory software verification for single-use disposable cassette controllers (`IEC 62304` safety class) prioritizes *deterministic execution without dynamic heap allocation (`malloc`), absence of non-deterministic floating-point library (`<math.h>`) or FPU exceptions, and statically bounded SRAM residency* portable to FPU-less safety microcontrollers and integer RISC-V cores (`RV32IM/Zve32x`). We therefore implemented the complete 8-state cassette plant, $\Sigma\Delta$ `RK4` predictor, 5-channel sequential Huber-EKF, integer NIS biofouling monitor, and acidosis bypass-flush state machine in freestanding integer C99 (`build/coralnpu/cart_bioreactor_hil_testbench.c`,

Table 5: **Freestanding FPU-Free C99 Fixed-Point Plant and 5-Channel Huber-EKF Verification.** Exact metrics emitted directly by `cart_bioreactor_hil_testbench.c --json`.

Firmware / Numerical Verification Metric	**IEEE-754** `FP64`	**Integer C99** (`Q16.16/Q12.20/Q8.24`)
Zero-Heap Static SRAM Workspace (`sizeof`)	–	2,704 B/8,192 B (33.01%)
Nominal Gate Switch Time (N = 1,000)	66.024 h	66.024 h
Nominal Fold Expansion N(168 h)	62.7979×	62.7802×
Nominal T_{SCM}(168 h) / h(168 h)	73.1989% / 6.7257%	73.1881% / 6.7253%
`RK4` ℓ_2 Error (N = 500 / 1,000 / 2,000)	Ref.	0.0085% / 0.0082% / 0.0078%
`RK4` ℓ_2 Error (N = 4,000 / 8,000)	Ref.	0.0114% / 0.0111%
Naive `Q16.16` Error (N = 1,000 / 8,000*)*	Ref.	0.6412% / 5.2906% *(drift)*
5-Ch Huber-EKF $\hat{h}$ RMSE (Biofouled Run)	2.3346%	2.3389% (0.0216% diff)
NIS > 13.5 Handover / Closed-Loop Gate	58.20 h / 84.00 h	58.20 h / 84.00 h
Acidosis Fault min pH (No Flush / Flush)	6.1366 / 6.9106	6.1366 / 6.9106

compiled under `-O2 -Wall -Wextra -Werror -std=c99` with zero unused neural-network weight tables).

States are stored in signed 32-bit `Q16.16` with dynamic pre-scaling ($N_{cells}/64$, $R_{LP}/32$, $S_{SRC}/8$, pH/8), `RK4` slopes in `Q8.24` ($2^{-24} \approx 5.96 \times 10^{-8}$) with 64-bit $\Sigma\Delta$ remainder tracking, and the 8×8 EKF covariance matrix $\mathbf{P}$ in signed `Q12.20` (spanning $[10^{-6}, 2048)$ so both initial R_{LP} variance 4.0 and optical `ORR` noise variance 1.44×10^{-4} are resolved without overflow or underflow). Sequential scalar measurement updates use an exact 64-bit integer square root (`isqrt64_u32`) for Huber innovation clipping ($|\nu_i| \le 2.5\sqrt{S_i}$) and an exact division-free integer biofouling check ($2\nu_1^2 > 27S_1 \iff \text{NIS}_1 > 13.5$). Table 5 summarizes exact metrics emitted by `cart_bioreactor_hil_testbench.c --json`:

1. **Zero-Heap Static SRAM Footprint:** Removing unused neural projection weights shrinks the static workspace (`coralnpu_cart_dtcm_bank0_workspace_t`) to 2,704 B (33.01% of an 8,192 B SRAM bank), encompassing all `Q16.16` state/checkpoint vectors, `Q12.20` 8×8 covariance and process-noise matrices, `Q8.24` `RK4`/$\Sigma\Delta$ registers, 64-frame persistence buffers, and stack margin.
2. **Deadband-Free `RK4` Parity Across Step Sizes:** At N = 1,000 steps, `Q16.16`+$\Sigma\Delta$ matches `FP64` switching time to the exact step (66.024 h), final expansion (62.7802× vs. 62.7979×), and T_{SCM} (73.1881% vs. 73.1989%), keeping normalized ℓ_2 trajectory error flat at 0.0078%–0.0114% across $N \in \{500, \ldots, 8{,}000\}$ (478× lower than naive `Q16.16`'s 5.2906% drift at N = 8,000).
3. **Full Integer 5-Channel Huber-EKF &** NIS **Handover Parity:** Across 560 closed-loop frames (168 h) on a mismatched, biofouled oncology donor (h_0 = 51.0%, α_{OD} = 1.05), the integer C99 5-channel Huber-EKF achieves $\hat{h}$ RMSE = 2.3389% vs. 2.3346% in IEEE-754 `FP64` (0.0216% **C99-vs.-`FP64` trajectory difference**), triggering $\text{NIS}_1 > 13.5$ optical handover (58.20 h) and closed-loop switching (84.00 h) at the exact same frames as `FP64`.

7 Limitations, Postulated Ex Vivo Falsification Blueprint, and Reproducibility

7.1 Model Limitations and Modeling Assumptions

Several limitations bound the interpretation of these computational results:

1. **Absence of Wet-Lab Execution:** No primary human T cells, CAR lentiviral vectors, or bioreactor

cassettes were experimentally cultured in this study. Rate constants tagged **[F]** in Table 1 represent phenomenologically calibrated couplings synthesized from separate immunological and structural reports [1, 3, 4, 7] rather than joint system-identification fits to a single time-series dataset.

2. **Deterministic Population-Mean Approximation:** The governing ODEs track mean intracellular organelle mass and population-level T_{SCM}/TOX^+ fractions, neglecting clone-to-clone CAR copy-number variance, $CD4^+$/$CD8^+$ subset interactions, and asymmetric mitochondrial partitioning at cytokinesis.
3. **Optical Redox Confounding at High Suspension Density:** While cell concentration cancels in first-order two-band autofluorescence `ORR` (5) [14], inner-filter reabsorption at ultra-high hollow-fiber densities ($> 3 \times 10^7$ cells/mL) and riboflavin/phenol-red media fluorescence require synchronized cell-free perfusate blanking in physical hardware.

7.2 Postulated Ex Vivo Benchtop Falsification Protocol

To enable direct experimental falsification without requiring *in vivo* animal studies, our simulations generate concrete, quantitative predictions testable in a 5-arm parallel perfusion bioreactor study using primary human CD19-BBζ CAR-T cells from $n \geq 20$ pretreated oncology donor apheresis samples:

- **Arm 1 (Standard Warburg Control):** 25 mM glucose + 300 IU/mL IL-2 (0–168 h). *Prediction:* High expansion ($> 70\times$), high $TMRE^{low}$/$MitoTracker^{high}$ depolarized fraction ($> 50\%$), $T_{SCM} < 10\%$.
- **Arm 2 (Continuous Urolithin A Monotherapy):** Media-A (25 μM urolithin A) continuously (0–168 h) in 25 mM glucose. *Prediction:* Clearance of $TMRE^{low}$ mitochondria ($< 5\%$) but incomplete T_{SCM} conversion ($\sim$ 25–35%) due to persistent glycolytic TOX^+ acquisition.
- **Arm 3 (Fixed 68 h Open-Loop Switch):** Media-A (0–68 h) switched blindly at 68.0 h to Media-B (6.5 mM glucose + 18.5 mM galactose, 1.0 μM `AZD7545`, 1.0 mM nicotinamide riboside, 10 ng/mL IL-7/IL-15).
- **Arm 4 (Chemical-Probe-Free `ORR`-EKF Closed-Loop + Adaptive 168–240 h Harvest):** Closed-loop `ORR`+OUR_{vol}+`OD`+`pH` Huber-EKF switching (6) from Media-A to STAC-free Media-B (`AZD7545` + `NR` only, avoiding `SRT1720` off-target risk [12]), with `OD`-triggered harvest up to 240 h for slow-dividing donors. *Falsification Criterion:* The model is falsified if Arm 4 fails to increase the 5-criteria pass rate ($N \geq 50\times$, $CD45RA^+CD62L^+TCF7^+ \geq 68\%$, $TOX^+ \leq 15\%$, Seahorse SRC $\geq 3.5\times$) relative to Arm 3, or if hypoproliferative ($\mu_{glyc} < 0.85\mu_{nom}$) donors fail to reach $\geq 50\times$ viable fold expansion during the 168–240 h consolidation window (Figure 3B).
- **Arm 5 (`SIRT1`-KO / `Glu230Lys` & Single-Target Ablation Controls):** Closed-loop switching omitting either `AZD7545` or `NR` (testing the single-target ablation collapse predicted in Table 3), plus `SIRT1` CRISPR knockout / `Glu230Lys` point-mutant specificity controls if synthetic STACs (`4TQ`/`SRT2104`) are evaluated [13].

7.3 Code and Reproducibility Availability (MIASE)

All structural audits, 8-state `FP64` simulations, 200-variable discrete-adjoint `L-BFGS-B` optimizations, Observability Gramians, $N = 1{,}000$ donor Monte Carlo runs and exact paired McNemar tests (PRNG seed `20261111`), publication vector figures, and freestanding C99 fixed-point verification benchmarks are deterministically regenerated from source via `simulator/generate_paper2_verified_benchmarks.py` and the freestanding C99 testbench `build/coralnpu/cart_bioreactor_hil_testbench.c` (repository: https://github.com/deesatzed/mitorestorer).

A Mathematical Appendix: Complete Bioreactor ODEs, Quotient Proof, Adjoint Sensitivities, and Sensor Models

A.1 Complete 8-State Perfusion Cassette ODEs and Closed-Form Auxiliary Rates

To guarantee zero discrepancy between analytical expressions, `eval_cart8_rhs_vec` in Python, and `eval_cart8_rhs_q16_to_q24` in C99, all ten intermediate bioenergetic rate functions are given in closed form:

$$g_{\mathrm{glc}}(u_2) = 1 - \alpha_{\mathrm{gal}}\, u_2, \qquad g_{\mathrm{pdh}}(u_2) = 0.15 + 0.85\, u_2, \tag{A.1}$$

$$v_{\mathrm{unmask}}(R_{\mathrm{LP}}) = \frac{K_{\mathrm{LP}}}{K_{\mathrm{LP}} + (R_{\mathrm{LP}} - R_{\mathrm{basal}})_+}, \tag{A.2}$$

$$k_{\mathrm{mito},D}(u_1, R_{\mathrm{LP}}) = k_{\mathrm{mito},W} + (k_{\mathrm{mito},D,\max} - k_{\mathrm{mito},W})\, u_1\, v_{\mathrm{unmask}}(R_{\mathrm{LP}}), \tag{A.3}$$

$$\mathrm{ATP}_{\mathrm{ox}}(\mathbf{x}, u_2) = (W + \eta_D D)(1 + \gamma_{\mathrm{pdh}} u_2)\big(1 + \gamma_{\mathrm{src}}(S_{\mathrm{SRC}} - 1)_+\big), \tag{A.4}$$

$$\mathrm{ATP}_{\mathrm{glyc}}(\mathbf{x}, \mathbf{u}) = \kappa_{\mathrm{glyc}}\, g_{\mathrm{glc}}(u_2)\big(1 - c_{\mathrm{akt}} u_1 (1 - u_2)\big)\frac{R_{\mathrm{LP}}}{R_{\mathrm{LP}} + K_{R,\mathrm{ATP}}}, \tag{A.5}$$

$$\phi_{\mathrm{pH}}(\mathrm{pH}) = \frac{1}{1 + \big((7.02 - \mathrm{pH})_+ / 0.16\big)^2}, \tag{A.6}$$

$$\dot{e}_{\mathrm{load}}(\mathbf{x}) = \frac{1 + \beta_{\mathrm{LP}}(R_{\mathrm{LP}} - R_{\mathrm{basal}})_+ / K_{\mathrm{LP}} + \beta_h h + \beta_{\mathrm{pH}}(7.02 - \mathrm{pH})_+}{\max(0.26,\ W + \eta_D D)}, \tag{A.7}$$

$$\mathrm{EP}(\mathbf{x}, \mathbf{u}) = \dot{e}_{\mathrm{load}}(\mathbf{x})\,(1 - c_1 u_1 + c_2 u_2)^2, \qquad d_{\mathrm{ros}}(\mathbf{x}, \mathbf{u}) = k_{\mathrm{dmg}}\big(\mathrm{EP}(\mathbf{x}, \mathbf{u}) - \theta_{\mathrm{GSH}}\big)_+, \tag{A.8}$$

$$w_{\mathrm{func}}(\mathbf{x}) = W(1 - h), \qquad \Phi_{\mathrm{ox}}(\mathbf{x}) = \frac{w_{\mathrm{func}}(\mathbf{x})^2}{w_{\mathrm{func}}(\mathbf{x})^2 + K_{W,\mathrm{div}}^2}. \tag{A.9}$$

Together with (1a)–(1c) in §2 and $\mathrm{ATP}_{\mathrm{tot}} = \mathrm{ATP}_{\mathrm{ox}} + \mathrm{ATP}_{\mathrm{glyc}}$, the remaining five state ODEs governing extracellular lactate-to-pyruvate balance, `FOXO1` nuclear shuttling, cristae spare respiratory capacity, TOX^+ epigenetic commitment, and perfusate pH are:

$$\begin{aligned}\dot{R}_{\mathrm{LP}} = {} & k_{\mathrm{Warburg}} g_{\mathrm{glc}}(u_2)\big(1 - 0.35 u_1(1 - u_2)\big)(1 + \alpha_h h) - k_{\mathrm{PDH}}(1 + \beta_{\mathrm{azd}} u_2) w_{\mathrm{frac}} R_{\mathrm{LP}} \\ & - (k_{\mathrm{perf,LP}} + k_{\mathrm{fl,LP}} u_{\mathrm{fl}})(R_{\mathrm{LP}} - R_{\mathrm{media}}),\end{aligned} \tag{A.10}$$

$$\begin{aligned}\dot{F}_{\mathrm{nuc}} = {} & k_{\mathrm{imp}}\big(f_0 + \alpha_{\mathrm{akt}} u_1 + \phi_{\mathrm{SIRT1}} u_2 g_{\mathrm{pdh}}(u_2)\big)\phi_{\mathrm{pH}}(\mathrm{pH})\,(1 - F_{\mathrm{nuc}}) \\ & - k_{\mathrm{exp}}(1 - 0.52 u_1)\big(1 - 0.85 u_2 g_{\mathrm{pdh}}(u_2)\big)\big(1 + \gamma_{\mathrm{acid}}(6.90 - \mathrm{pH})_+\big) F_{\mathrm{nuc}},\end{aligned} \tag{A.11}$$

$$\dot{S}_{\mathrm{SRC}} = s_{\mathrm{basal}}(W + \eta_D D) - k_{\mathrm{src,dec}}\, S_{\mathrm{SRC}} + k_{\mathrm{cristae}} w_{\mathrm{func}}(\mathbf{x})\big(1 + \beta_{\mathrm{src}} u_2 F_{\mathrm{nuc}} g_{\mathrm{pdh}}(u_2)\big), \tag{A.12}$$

$$\begin{aligned}\dot{E}_{\mathrm{TOX}} = {} & \Big[k_{\mathrm{exh}}(\mathrm{EP} - \theta_{\mathrm{exh}})_+ + k_{\mathrm{acid,exh}}(6.86 - \mathrm{pH})_+ + k_{\mathrm{warb,exh}} g_{\mathrm{glc}}(u_2)(1 - 0.22 u_1)\Big(1 + 0.70\frac{N_{\mathrm{cells}}}{N_{\mathrm{ref}}}\Big)\Big](1 - E_{\mathrm{TOX}}) \\ & - \big(r_{\mathrm{epig},0} + k_{\mathrm{rev,epig}} u_2 F_{\mathrm{nuc}} g_{\mathrm{pdh}}(u_2)\big) E_{\mathrm{TOX}},\end{aligned} \tag{A.13}$$

$$\dot{\mathrm{pH}} = (k_{\mathrm{perf},0} + k_{\mathrm{perf,fl}} u_{\mathrm{fl}})(\mathrm{pH}_{\mathrm{media}} - \mathrm{pH}) - k_{\mathrm{acid}} g_{\mathrm{glc}}(u_2)(R_{\mathrm{LP}} - R_{\mathrm{buf}})_+ \frac{N_{\mathrm{cells}}}{N_{\mathrm{ref}}}. \tag{A.14}$$

A.2 Proof of Proposition 1 (Stage-1 Quotient Selection)

Proof. Under Stage-1 actuation ($u_1 = 1, u_2 = 0$) with $\mathrm{EP}(\mathbf{x}, \mathbf{u}) \le \theta_{\mathrm{GSH}}$, we have $d_{\mathrm{ros}}(\mathbf{x}, \mathbf{u}) = 0$, so (1a)–(1b) reduce to

$$\dot{W} = \left[k_{\mathrm{bio},0}\left(1 - \frac{M}{K}\right)_+ - k_{\mathrm{mito},W}\right]W, \qquad \dot{D} = \left[k_{\mathrm{rep,adv}}k_{\mathrm{bio},0}\left(1 - \frac{M}{K}\right)_+ - k_{\mathrm{mito},D}(1, R_{\mathrm{LP}})\right]D. \tag{A.15}$$

Applying the quotient identity $\dot{h} = (\dot{D}W - D\dot{W})/M^2$ with $W/M = 1 - h$ and $D/M = h$ gives

$$\begin{aligned}\dot{h} &= \frac{DW}{M^2}\left[k_{\mathrm{rep,adv}}k_{\mathrm{bio},0}\left(1 - \frac{M}{K}\right)_+ - k_{\mathrm{mito},D}(1, R_{\mathrm{LP}}) - k_{\mathrm{bio},0}\left(1 - \frac{M}{K}\right)_+ + k_{\mathrm{mito},W}\right] \\ &= -\lambda_{\mathrm{CART}}(t)\, h(1 - h),\end{aligned} \tag{A.16}$$

establishing (3). Separating variables $dh/[h(1 - h)] = -\lambda_{\mathrm{CART}}(t)\, dt$ and integrating from 0 to t yields $\ln[h(t)(1 - h_0)/(h_0(1 - h(t)))] = -s(t)$, which inverts directly to $h(s) = [1 + ((1 - h_0)/h_0)e^s]^{-1}$ and $s^* = \ln[h_0(1 - h_{\mathrm{gate}})/(h_{\mathrm{gate}}(1 - h_0))]$. □

A.3 Discrete-Adjoint `RK4` Sensitivity Recursion

To render the discrete-time `RK4` transition map $\mathbf{x}_{k+1} = \Phi_k(\mathbf{x}_k, \mathbf{u}_k)$ continuously differentiable for `L-BFGS-B` optimization in (4), nonsmooth positive-part hinges $(z)_+$ are replaced during gradient evaluation by the C^∞ hyperbolic approximation

$$\max_\epsilon(0, z) = \frac{1}{2}\left(z + \sqrt{z^2 + \epsilon^2}\right), \qquad \frac{d}{dz}\max_\epsilon(0, z) = \frac{1}{2}\left(1 + \frac{z}{\sqrt{z^2 + \epsilon^2}}\right), \tag{A.17}$$

with $\epsilon = 10^{-3}$ inside the ODE right-hand side and $\epsilon = 0.25$ on the terminal 50× dose penalty in (4) (with final reported protocol metrics always audited on the exact nonsmooth $\epsilon = 0$ plant at $N = 1{,}000$ fine steps). Initializing the terminal costate $\mathbf{p}_{N_{\mathrm{ocp}}} = \nabla_{\mathbf{x}(T)}J_\alpha(\mathbf{u}) \in \mathbb{R}^8$, exact discrete-time sensitivities are propagated backward through all four internal `RK4` stages at each step $k = N_{\mathrm{ocp}} - 1, \ldots, 0$ via

$$\mathbf{p}_k = \left(\frac{\partial\Phi_k(\mathbf{x}_k, \mathbf{u}_k)}{\partial\mathbf{x}_k}\right)^\top \mathbf{p}_{k+1}, \qquad \frac{\partial J_\alpha}{\partial\mathbf{u}_k} = \frac{0.016}{N_{\mathrm{ocp}}}\mathbf{u}_k + \left(\frac{\partial\Phi_k(\mathbf{x}_k, \mathbf{u}_k)}{\partial\mathbf{u}_k}\right)^\top \mathbf{p}_{k+1}. \tag{A.18}$$

A.4 Physical 5-Channel Sensor Equations and Closed-Loop Reachability Gate

Sampled every $\Delta t_{\mathrm{obs}} = 0.30$ h under correlated AR(1) sensor noise $\nu_{i,k} = \rho\nu_{i,k-1} + \sqrt{1 - \rho^2}\,\sigma_{i,k}\varepsilon_{i,k}$ ($\rho = 0.65$, $\varepsilon_{i,k} \sim \mathcal{N}(0, 1)$), the five physical cassette channels are:

$$y_{1,k} = R_{\mathrm{LP}}(t_k) + b_{\mathrm{foul}}(t_k) + \nu_{1,k}, \qquad \sigma_1 = 0.55, \tag{A.19a}$$

$$y_{2,k} = N_{\mathrm{cells}}(t_k)\,\mathrm{ATP}_{\mathrm{ox}}(\mathbf{x}(t_k), u_2) + \nu_{2,k}, \qquad \sigma_{2,k} = 0.030\sqrt{1 + \left(0.08\, N_{\mathrm{cells}}(t_k)\right)^2}, \tag{A.19b}$$

$$y_{3,k} = \alpha_{\mathrm{OD}}\, N_{\mathrm{cells}}(t_k) + \nu_{3,k}, \qquad \sigma_{3,k} = \sqrt{0.35^2 + \left(0.025\, N_{\mathrm{cells}}(t_k)\right)^2}, \tag{A.19c}$$

$$y_{4,k} = \mathrm{pH}(t_k) + \nu_{4,k}, \qquad \sigma_4 = 0.018, \tag{A.19d}$$

$$y_{5,k} = \frac{W(t_k)}{W(t_k) + D(t_k) + c_{\mathrm{glyc}}\, g_{\mathrm{glc}}(u_2)} + \nu_{5,k}, \qquad \sigma_5 = 0.012. \tag{A.19e}$$

In the closed-loop switching law (6), the online-estimated relative donor proliferation rate $\hat{\mu}_{\mathrm{rel}}(t)$, predicted Stage-2 expansion factor $M_{\mathrm{S2}}(t)$, and reachability threshold $N_{\mathrm{req}}(t, \hat{\mu}_{\mathrm{rel}})$ are defined explicitly by

$$\hat{\mu}_{\mathrm{rel}}(t) = \begin{cases} 1.0, & t < 28 \text{ h}, \\ \mathrm{clip}\left(\dfrac{\ln\big(\hat{N}_{\mathrm{cells}}(t)/\hat{N}_{\mathrm{cells}}(24\text{ h})\big)}{0.044\,(t-24)},\ 0.78,\ 1.22 \right), & t \geq 28 \text{ h}, \end{cases} \tag{A.20}$$

$$M_{\mathrm{S2}}(t) = \exp\Big(0.0122\sqrt{\hat{\mu}_{\mathrm{rel}}(t)}\,(T_{\mathrm{plan}} - t)_+\Big), \tag{A.21}$$

$$N_{\mathrm{req}}(t, \hat{\mu}_{\mathrm{rel}}) = \mathrm{clip}\left(\frac{N_{\mathrm{target}}}{\max\big(1.05,\ 0.58\,M_{\mathrm{S2}}(t) + 0.42\min(M_{\mathrm{S2}}(t),\ 3.6)\big)},\ 14.0,\ 42.0 \right), \tag{A.22}$$

with $T_{\mathrm{plan}} = 168$ h, $N_{\mathrm{target}} = 52.5\times$, and $N_{\mathrm{guard}} = 18.0\times$ for fixed 168-hour runs (and $T_{\mathrm{plan}} = 240$ h, $N_{\mathrm{target}} = 53.5\times$, $N_{\mathrm{guard}} = 14.0\times$ under the adaptive 168–240 h policy).

Acknowledgments and Disclosures

Acknowledgments & Program Support: This research was supported by Google LLC through the **Google AI for Science Researcher Access Program**, providing access to Google DeepMind's **Antigravity** agentic research environment, Gemini frontier reasoning models, and integrated structural biology and AlphaFold Database tools.

Author Contributions & AI Collaboration Disclosure: This study was conceived, directed, and conducted solely by Wayne A. Satz, MD in direct collaboration with Google DeepMind's Antigravity AI research agent; no other human investigators participated. Antigravity assisted with structural bioinformatics coordinate audits, mathematical derivation checks, Python discrete-adjoint and EKF cohort simulations, freestanding C99 fixed-point firmware implementation, and LaTeX manuscript preparation. Wayne A. Satz, MD audited, verified, and takes full scholarly responsibility for the integrity of all equations, code artifacts, numerical benchmarks, and scientific conclusions.

Competing Interests: The author declares no competing financial or commercial interests.

Ethics Statement: This study consists exclusively of theoretical modeling, structural bioinformatics on public Protein Data Bank / AlphaFold entries, numerical ODE simulation, and host-emulated firmware verification; no human participants, primary donor tissues, clinical cohorts, or live animals were involved.

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